

Assignment 3 - EE4725 Quasi Optical Systems

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1 System Parameters

The system under analysis is an elliptical dielectric lens antenna with the following parameters:

- Operating frequency: $f = 200$ GHz, wavelength $\lambda_0 = c/f = 1.5$ mm, wavenumber $k_0 = 2\pi/\lambda_0$
- Dielectric material: fused silica, $\varepsilon_r = 4$, wavenumber in dielectric $k_d = \sqrt{\varepsilon_r} k_0 = 2k_0$
- Feed: planar antenna at the lower focal point of an elliptical lens, with Gaussian far field in infinite dielectric:

$$\vec{E}_{far}^a(\theta, \phi) = e^{-\left[\left(\frac{u}{u_0}\right)^2 + \left(\frac{v}{v_0}\right)^2\right]} \left(\sin \phi \hat{\theta} + \cos \phi \hat{\phi}\right) \frac{e^{-jk_d r}}{r} \quad (1)$$

where $u = \sin \theta \cos \phi$, $v = \sin \theta \sin \phi$, and u_0, v_0 control directivity in the principal planes.

- Elliptical lens geometry: eccentricity $e = 1/\sqrt{\varepsilon_r} = 0.5$, polar equation from lower focus:

$$r(\theta) = \frac{a(1 - e^2)}{1 - e \cos \theta} \quad (2)$$

with semi-major axis a , semi-minor axis $b = a\sqrt{1 - e^2}$, and center-to-focus distance $c = ea$.

2 Q1: Feed Far Field

The Gaussian feed pattern (1) with $u_0 = v_0 = 0.5$ is evaluated in the two principal planes:

- **E-plane** ($\phi = 90^\circ$): $|\vec{E}| = \exp(-\sin^2 \theta / v_0^2)$, purely $\hat{\theta}$ -polarized
- **H-plane** ($\phi = 0^\circ$): $|\vec{E}| = \exp(-\sin^2 \theta / u_0^2)$, purely $\hat{\phi}$ -polarized

Since $u_0 = v_0$, both planes yield identical patterns. The -3 dB half-angle is 24.6° and the -10 dB half-angle is 49.4° .

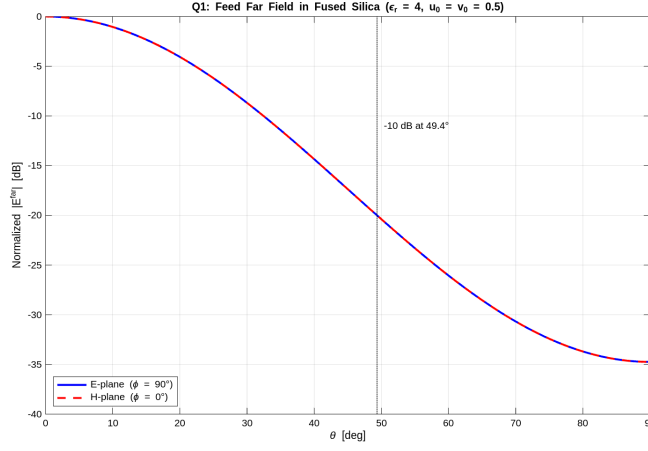


Figure 1: Normalized far-field pattern of the Gaussian feed in infinite fused silica ($\varepsilon_r = 4$, $u_0 = v_0 = 0.5$, 200 GHz). E-plane and H-plane curves overlap due to the circular symmetry $u_0 = v_0$.

Observations from Figure 1:

- The pattern is a smooth Gaussian roll-off with no sidelobes. The E-plane and H-plane curves are identical, confirming the circular symmetry of the feed for $u_0 = v_0$.
- The polarization ($\sin \phi \hat{\theta} + \cos \phi \hat{\phi}$) is a Huygens-type source: in the E-plane it is purely $\hat{\theta}$ -polarized, in the H-plane purely $\hat{\phi}$ -polarized. This ensures the co-pol pattern is the same in both planes.
- This circular symmetry will be broken when the lens is introduced (Q4, Q5), because the Fresnel transmission coefficients differ for TE and TM polarizations.

3 Q2: Fresnel Transmission Coefficients — Flat Interface

The Fresnel transmission coefficients for a flat fused silica-to-air interface are:

$$\tau^\perp = \frac{2\zeta_0 \cos \theta_i}{\zeta_0 \cos \theta_i + \zeta_d \cos \theta_t}, \quad \tau^\parallel = \frac{2\zeta_0 \cos \theta_i}{\zeta_0 \cos \theta_t + \zeta_d \cos \theta_i} \quad (3)$$

where $\zeta_d = \zeta_0/\sqrt{\varepsilon_r}$ and Snell's law gives $\sqrt{\varepsilon_r} \sin \theta_i = \sin \theta_t$. The power transmission ratio is:

$$\frac{P_t}{P_i} = |\tau|^2 \frac{\zeta_d \cos \theta_t}{\zeta_0 \cos \theta_i} = 1 - |\Gamma|^2 \quad (4)$$

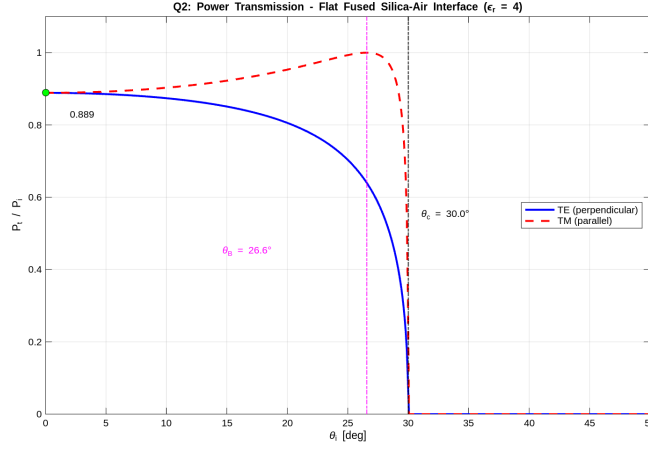


Figure 2: Power transmission ratio P_t/P_i vs. incidence angle for a flat fused silica–air interface ($\varepsilon_r = 4$), for TE and TM polarizations.

Observations from Figure 2:

1. **Normal incidence** ($\theta_i = 0^\circ$): $P_t/P_i = 4n/(1+n)^2 = 8/9 \approx 0.889$ for both TE and TM ($n = \sqrt{\varepsilon_r} = 2$). About 11.1% of power is reflected.
2. **Brewster angle** (TM only): $\theta_B = \arctan(1/\sqrt{\varepsilon_r}) = 26.6^\circ$. At this angle $\Gamma^\parallel = 0$ and TM transmission is complete ($P_t/P_i = 1$). No corresponding angle exists for TE, since $\Gamma^\perp$ increases monotonically with θ_i .
3. **Critical angle**: $\theta_c = \arcsin(1/\sqrt{\varepsilon_r}) = 30.0^\circ$. Beyond this angle, total internal reflection occurs for both polarizations: $\sin \theta_t > 1$, θ_t becomes complex, and all power is reflected.
4. **Between Brewster and critical** (TM): P_t/P_i drops from 1.0 back to 0 over a narrow 3.4° range, demonstrating the rapid transition to total internal reflection.

4 Q3: Fresnel Transmission Coefficients — Elliptical Lens ($D = 9\lambda_0$)

On the elliptical lens surface, the local incidence angle depends on the elevation angle θ from the lower focus via:

$$\cos \theta_i = \frac{1 - e \cos \theta}{\sqrt{1 + e^2 - 2e \cos \theta}} \quad (5)$$

The computed lens parameters for $D = 9\lambda_0 = 13.5\text{ mm}$ are: $e = 0.500$, $a = 7.794\text{ mm}$, $b = 6.750\text{ mm}$, lens height $h = a(1 + e) = 11.691\text{ mm}$, and maximum lens angle $\theta_{\max} = \arctan(b/(ea)) = 60.0^\circ$.

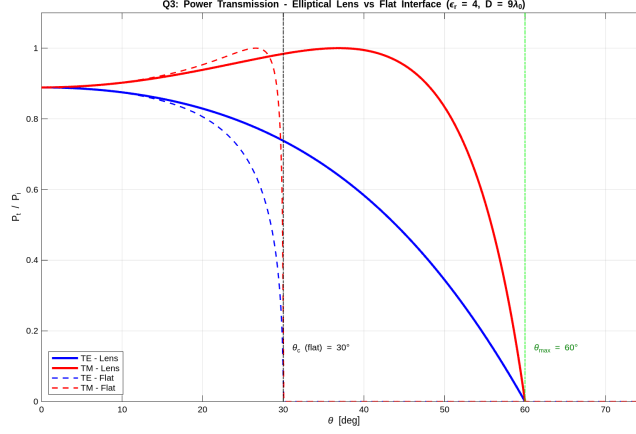


Figure 3: Power transmission ratio vs. elevation angle θ for the elliptical lens (solid) compared with the flat interface (dashed). The lens extends significant transmission well beyond the flat-interface critical angle of 30° .

Observations from Figure 3:

- At the top of the lens ($\theta = 0$), $\theta_i = 0^\circ$ and the transmission equals the flat normal-incidence value of 0.889 for both polarizations.
- The elliptical curvature causes the local incidence angle to grow much more slowly than θ itself. At $\theta = 40^\circ$ (the truncation angle used in Q4-Q6), the local $\theta_i \approx 17^\circ$, still well below the critical angle.
- The flat interface (dashed) shows total internal reflection at $\theta_i = 30^\circ$, but the lens maintains significant transmission well beyond $\theta = 30^\circ$. TM transmission (solid red) stays above 0.9 up to $\theta \approx 50^\circ$ due to the Brewster effect on the curved surface.
- At $\theta_{\max} = 60^\circ$, the local incidence angle reaches exactly 30° (the critical angle), and both polarizations drop to zero. This is a fundamental property of the elliptical lens with $e = 1/\sqrt{\epsilon_r}$: the maximum lens angle always maps to the critical angle.
- The key advantage of the elliptical geometry is clear: it extends the range of high transmission to much larger elevation angles compared to a flat interface, enabling wider feeds while maintaining good power transfer.

5 Q4: Aperture Current Distribution

The lens is truncated at $\theta_0 = 40^\circ$ with $\varepsilon_r = 4$, $D = 9\lambda_0$, and $u_0 = v_0 = 0.5$. The equivalent aperture currents are computed using the GO/PO method. The elliptical lens provides a spreading factor $S = 1$ (exact) and uniform phase $k_d r + k_0 r' = k_d a(1 + e)$ across the aperture.

Since the equivalent electric and magnetic surface currents are orthogonal with the same distribution, the far field can be computed using only $\vec{J}_s$ with twice the amplitude:

$$J_{s,x} = -\frac{2}{\zeta_0} \frac{1}{r} \left[\tau^\parallel E_i^\theta \cos \phi - \tau^\perp E_i^\phi \sin \phi \right] \quad (6)$$

$$J_{s,y} = -\frac{2}{\zeta_0} \frac{1}{r} \left[\tau^\parallel E_i^\theta \sin \phi + \tau^\perp E_i^\phi \cos \phi \right] \quad (7)$$

For the Huygens feed, $E_i^\theta = \sin \phi \cdot G(\theta)$ and $E_i^\phi = \cos \phi \cdot G(\theta)$, where $G(\theta)$ is the Gaussian envelope. In the principal planes this simplifies to:

- E-plane ($\phi = 90^\circ$): $J_y \propto \tau^\parallel \cdot G(\theta)/r$, $J_x = 0$
- H-plane ($\phi = 0^\circ$): $J_y \propto \tau^\perp \cdot G(\theta)/r$, $J_x = 0$

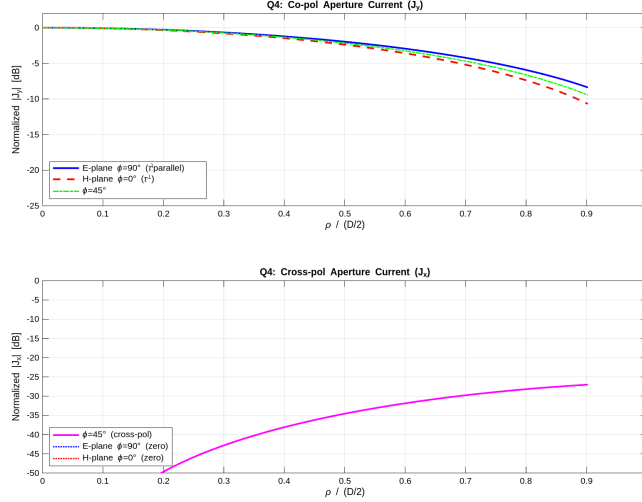


Figure 4: Normalized aperture current distribution. Top: co-pol component J_y in the E-plane ($\tau^\parallel$), H-plane ($\tau^\perp$), and $\phi = 45^\circ$ cut. Bottom: cross-pol component J_x at $\phi = 45^\circ$; J_x is identically zero in the principal planes.

Observations from Figure 4:

- **Co-pol** (J_y): The E-plane (governed by $\tau^\parallel$, TM) shows an edge taper of -8.4 dB, while the H-plane (governed by $\tau^\perp$, TE) shows a steeper taper of -10.7 dB. The TM coefficient benefits from the Brewster effect, maintaining higher transmission at oblique angles, hence the gentler taper in the E-plane.
- **Cross-pol** (J_x): In the principal planes ($\phi = 0^\circ$ and 90°), the cross-pol is identically zero because the Huygens polarization aligns perfectly with the TE/TM decomposition axes. At $\phi = 45^\circ$, cross-pol appears at -27.0 dB below co-pol, arising from the mismatch $\tau^\parallel \neq \tau^\perp$.
- **Effect of feed pattern** (Q1): The Gaussian taper dominates the amplitude variation. At $\theta_0 = 40^\circ$, the feed itself is at about -5 dB, which combined with the Fresnel coefficient variation and $1/r$ geometric taper produces the total 8–11 dB edge taper.
- **Effect of transmission coefficients**: Without Fresnel effects ($\tau^\perp = \tau^\parallel$), the aperture distribution would be perfectly circularly symmetric. The E/H asymmetry and the cross-pol generation are entirely due to $\tau^\perp \neq \tau^\parallel$.

6 Q5: Lens Far Field

The lens far field is computed via the free-space spectral Green's function applied to the aperture currents from Q4:

$$\vec{E}_{lens}^{far}(\vec{r}) = jk_{zs} \tilde{G}_{fs}^{ej}(k_{xs}, k_{ys}) \tilde{J}_s(k_{xs}, k_{ys}) \frac{e^{-jk_0 r}}{2\pi r} \quad (8)$$

where $\tilde{J}_s$ is the 2D Fourier transform of the aperture currents, evaluated at the stationary phase point $(k_{xs}, k_{ys}) = k_0(\sin \theta \cos \phi, \sin \theta \sin \phi)$. The computation uses a 301×301 aperture grid with 1001 far-field angle samples.

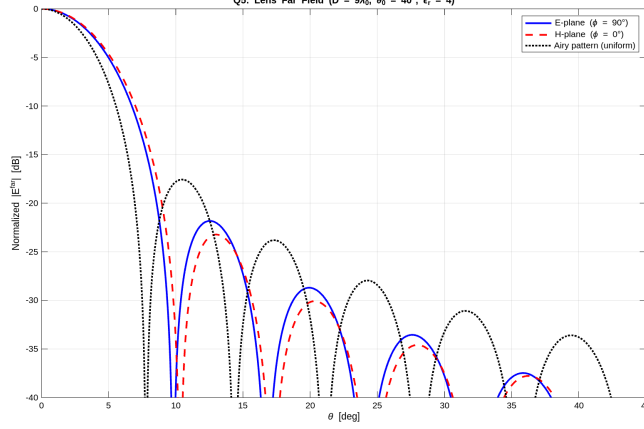


Figure 5: Normalized lens far-field pattern in the E-plane and H-plane, compared with the Airy pattern of a uniformly illuminated circular aperture of diameter $D = 9\lambda_0$.

Observations from Figure 5:

1. **Wider main beam:** The first null of the lens pattern is at $\sim 9^\circ$ – 10° , compared to $\sim 8^\circ$ for the Airy pattern. The Gaussian + Fresnel amplitude taper reduces the effective aperture size, broadening the main beam.
2. **Lower sidelobes:** The first sidelobe is approximately -22 dB (E-plane) and -26 dB (H-plane), well below the Airy first sidelobe at -17.6 dB. The smooth taper acts as a window function that suppresses sidelobes at the cost of a wider main beam.
3. **E/H plane asymmetry:** The H-plane shows deeper nulls and lower sidelobes than the E-plane. This asymmetry is governed by:
 - *Primary cause:* $\tau^\perp \neq \tau^\parallel$. The E-plane co-pol uses $\tau^\parallel$ (TM, gentler taper $\rightarrow$ narrower beam, higher sidelobes), while the H-plane uses $\tau^\perp$ (TE, steeper taper $\rightarrow$ wider beam, lower sidelobes). This is consistent with the aperture current asymmetry observed in Q4 (Figure 4).
 - *Secondary cause:* The free-space SGF introduces a $\cos\theta$ factor difference between E_θ and E_ϕ projections.

7 Q6: Directivity and Gain

The directivity is computed from the far-field radiation intensity integrated over the upper hemisphere on a 301×181 angular grid:

$$D = \frac{4\pi U_{\max}}{P_{\text{rad}}}, \quad U(\theta, \phi) = |E_\theta|^2 + |E_\phi|^2 \quad (9)$$

The gain accounts for power lost to spillover and Fresnel reflection:

$$G = D \cdot \eta_r, \quad \eta_r = \frac{P_{rad}^{lens}}{P_{rad}^{feed}} \quad (10)$$

where η_r includes both spillover losses (feed power missing the lens) and Fresnel reflection losses at the lens surface.

Table 1: Directivity, gain, and efficiency breakdown for the lens antenna of Q4.

Parameter	Value
$D_{\max}$ (uniform aperture, $(\pi D/\lambda_0)^2$)	29.0 dBi
Directivity D	27.8 dBi
Gain $G = D \cdot \eta_r$	27.0 dBi
Spillover efficiency η_s	94.8%
Reflection efficiency η_{refl}	87.8%
Total $\eta_r = \eta_s \cdot \eta_{\text{refl}}$	83.3%
Aperture efficiency $D/D_{\max}$	74.9%

The results in Table 1 can be interpreted as follows:

- The directivity is 1.2 dB below $D_{\max}$, due to the amplitude taper from the Gaussian feed pattern and the Fresnel coefficient variation across the aperture.
- The gain is a further 0.8 dB below the directivity, due to the combined spillover and reflection losses.
- The high spillover efficiency (94.8%) is expected: the feed -10 dB angle (49.4°) is well beyond the lens truncation angle (40°), so most feed power is intercepted by the lens.
- The reflection efficiency (87.8%) represents the average Fresnel loss across the lens surface, which is moderate for fused silica ($\varepsilon_r = 4$).
- The overall aperture efficiency of 74.9% is a good result for an unmatched dielectric lens antenna, and could be improved further with a quarter-wave matching layer ($\varepsilon_{r,m} = \sqrt{\varepsilon_r} = 2$, thickness $\lambda_m/4$), which would eliminate the broadside reflection and improve η_{refl} .